MATTHEW CASTELLO

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EDUCATION

University of California, San Diego *Sept 2026 – Present*
B.S. Computer Science (in progress)
Relevant coursework: CSE 11 (Intro to Programming), CSE 20 (Discrete Math), MATH 18 (Linear Algebra), LIGN 5

Buchanan High School — Clovis, CA *Graduated June 2026*
GPA: 4.0 unweighted / 4.45 weighted | Concurrent enrollment at CSU Fresno and Clovis Community | 3-year CTE Engineering Pathway

TECHNICAL EXPERIENCE

Junior Programmer, PreLicenseTraining (Remote) *April 2026 - Sep 2026*
- Built backend features in PHP for an online insurance exam-prep platform, including three interactive study games (crossword, true/false, trivia-show format) as alternatives to flashcard review.
- Ran ongoing QA testing to find and fix site bugs; worked about 170 hours over the summer.

Administrative Lead, Buchanan Bird Brains (*FIRST* Robotics Competition) *Aug 2025 – June 2026*
- Organized team events and outreach and managed onboarding and leadership nominations for a 80+ member team.

Programming Director, Buchanan Bird Brains (*FIRST* Robotics Competition) Aug *2023 – June 2025*
- Led a subteam of 4–11 student programmers and built an onboarding curriculum that grew the number of competition-ready programmers from 2 to 8 over two years.
- Contributed to software and strategy changes that helped the team climb 350+ places in competitive ranking and qualify for the *FIRST* World Championship two years in a row.
- Designed a constraint-programming scheduling system that generates fair, optimized schedules for 80+ team members in under 15 seconds, replacing a manual process prone to unfair breaks and repeat pairings.
- Built an Android scouting app (Java/XML) that encodes match data into QR codes for offline collection at Wi-Fi-restricted competitions, auto-syncing to Google Sheets for alliance-selection analysis; later rebuilt it as a cellular-based scouting website with an interactive field-diagram tool for drawing robot paths.

PERSONAL PROJECTS

Jarvis — Local AI Voice/Text Assistant (Python)
- Built a locally hosted assistant combining whisper.cpp speech-to-text with an Ollama/Qwen3 LLM and a tiered permission system that gates read-only, logged, and confirmation-required actions to keep the model from taking unsafe system actions.

Wallpaper Engine — Linux Desktop Controller (Python)
- Built a multi-monitor animated wallpaper manager for Linux (X11) using mpv/xwinwrap, with occlusion detection that pauses playback when windows are fully covered to save CPU/GPU usage.

LEADERSHIP & VOLUNTEER WORK

- Teens That Care (TTC), Leadership Council: 350+ hours; organized a 7-week summer camp for at-risk children, growing weekly attendance to 30 participants. Served at homeless shelters, adaptive sports events, food banks, etc.
- Make-A-Wish: 170+ hours of event planning and fundraising that helped grant 3 wishes to critically ill children.
- *FIRST* LEGO League: coached two elementary-level robotics teams in programming and design fundamentals; judged and refereed six regional competitions.
- Tutoring Club, VP: organized and led peer tutoring, expanding reach across multiple classes.

AWARDS & HONORS

- *FIRST* Impact Award, FRC Regional Finalist (2x), FRC Quality Award (2x); Mayoral Certificate of Recognition (2x);
- Silver President's Volunteer Service Award; Silver National Service Honor Award;
- Community Hero Award (top out of 145+ members)

SKILLS

Languages: Java, Python, PHP (basic), JavaScript, HTML/CSS
Tools & Platforms: Git/GitHub, VS Code, Linux (daily drive Mint), Figma, Google Sheets API
Concepts: ML fundamentals, autonomous pathfinding, PID control, computer vision basics